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Electromyographical indication of muscular fatigue in occupational field studies

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Abstract

Surface electromyography offers a valuable tool for the indication of muscular fatigue in occupational field studies. For this purpose, the time course of the electromyogram (EMG) has to be analysed, in order to detect typical fatigue-induced changes such as an increase in EMG amplitude or a shift in the spectral distribution towards lower frequencies. Such procedures need a detailed knowledge about the actual activity of the person and muscle under test for the total working time. This can be attained by encoding the activity of the person during the work and recording an electrical code signal together with the electrophysiological signals. For the indication of muscle fatigue, EMGs for situations connected with similar muscle load should only be compared, since the EMG amplitude as well as the spectrum do not only depend upon the fatigue state, but also upon the produced muscle force. This demand can be fulfilled by (i) interrupting the work and performing test contractions of known force in a predefined body posture or (ii) by comparing situations connected with a certain reference activity. (iii) In a recently developed approach for the joint analysis of EMG amplitude and spectrum (JASA) changes in the amplitude and the spectrum are considered simultaneously. This method permits the discrimination between fatigue-induced and force-related EMG changes. Using this procedure, changes in the EMG can be attributed to categories like fatigue or recovery as well as increase or decrease in the force production of the muscle under test. Applications from field studies during manual materials handling in a weaving mill, price recording at scanner checkouts in a supermarket and the performance of surgical work using endoscopic operation techniques in urology demonstrate the appropriateness of electromyography for fatigue indication in occupational physiology and ergonomics. Nevertheless, the commonly used measures of muscular fatigue such as increase in EMG amplitude and left shift in EMG spectrum are primarily related to the electrical activation and its propagation along the muscle fibres. Their connection to the fatigue-induced reduction in the force generating capacity of the muscle under test includes complex physiological implications. Therefore, the need for further development of fatigue indicators which are more directly related to muscular force is recognized.

Relevance to industry

In occupational health and ergonomics indication of muscular fatigue is needed, since activities inducing muscular fatigue can be performed for a limited time, only, and the quality of work can be influenced negatively. Electromyography offers valuable tools for the indication of fatigue and the appropriate assessment of ergonomic design measures. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Muscular fatigue; Electromyography; EMG amplitude; EMG spectrum; Fatigue indication; Occupational field studies

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1. Introduction

During sustained or repetitive muscle contractions typical changes in surface electromyograms (EMG), such as an increase in the amplitude or a shift in the frequency spectrum towards lower frequencies, can be observed. In occupational electromyography such EMG changes are commonly interpreted as signs of muscular fatigue and used to establish the occurance of fatigue. Since, however, the EMG amplitude as well as its spectral content do not only depend on the fatigue state, but also on the force production of the muscle under test, changes in the amplitude and spectrum cannot be unequivocally attributed to muscular fatigue.

Under laboratory conditions it is normally possible to control the force produced by the muscle under test. Under such circumstances the force can be kept constant at a known level and a change in the respective EMG can be ascribed to a change in the fatigue state of the muscle. Under field conditions, however, the force production is determined by the actual necessities of the work and cannot be controlled by the investigator. In a general case, it is therefore not possible to decide whether a temporal variation of an EMG is caused by a change in the force production or in the fatigue state. Nevertheless, the indication of muscular fatigue is possible under shop-floor conditions if special requirements are satisfied.

One possible way is to compare the EMGs for situations in which the muscle force is identical. Two methods which fulfil this demand were previously applied in occupational studies: (i) the execution of test contractions of known force in a defined posture, and (ii) the comparison of EMGs for reference activities of comparable workload. In another approach (iii) amplitude and spectral shifts are considered simultaneously in order to discriminate between force-related and fatigue-induced EMG changes. To this end a method for the "joint analysis of EMG spectrum and amplitude (JASA)" was developed and applied (Luttmann et al., 1996a,b; Luttmann and Jäger, 1998). Using this procedure time-related EMG changes can be assigned to different categories such as fatigue, recovery, force increase, or force decrease.

The aim of this paper is to present several methods for the indication of muscular fatigue and to weigh their appropriateness using examples from various shop-floor studies.

2. Methodology for the identification of muscular fatigue in occupational field studies

2.1. Preliminary remarks

In the context of ergonomics and work-physiology muscle fatigue is understood as a reduction in the force generating capacity of a muscle which occurs in the course of the work. A direct fatigue determination based on the mechanical performance capacity of a muscle needs the repeated execution of maximum voluntary contractions at several moments of time during the work. Such a procedure should not be applied in real work situations, since the measuring strongly disturbs the work flow. Furthermore, the measuring is retroactive in its nature, because the execution of test contractions reduces the performance capacity. Therefore, in occupational studies an indirect fatigue determination using electromyography is preferred. All electromyographical fatigue analyses are based on the assumption that the change in the mechanical performance capacity is reflected in corresponding changes in the myoelectrical signal of the muscle under test. Many studies performed since the beginning of this century (e.g. Piper, 1909,1912; Cobb and Forbes, 1923) have shown that fatiguing contractions of a muscle coincide with typical changes in the EMG time course, like an increase in the EMG amplitude or a shift in the EMG spectrum towards lower frequencies. In order to detect those changes, long-term EMG recordings of selected muscles or muscle groups are performed for certain periods of time up to total working shifts or considerable shift sections. In data evaluation it has to be examined whether typical fatigue-induced changes in the EMG can be found.

The preferable method for the measurement of EMGs in occupational settings is surface electromyography, since this method allows a noninvasive recording of myoelectrical signals and a minimal

restraint of the subjects. The EMGs can be stored using portable recorders or the signal can be transmitted using a portable telemetric device. Especially for the use of portable recorders, data reduction is sometimes advisable. For this purpose characteristics representing the averaged amplitude like Root Mean Square (RMS) and Electrical Activity (EA) or spectral parameters like Mean Power Frequency (MPF) and Median Frequency (MF) are calculated from the raw EMGs. In field applications the recording of raw EMGs should be preferred, since recordings subjected to prior data reduction have the disadvantage that disturbances caused, for example, by strong electrode movements or power-line hum cannot be distinguished from the electrical signal originating at the muscle fibres.

2.2. Test contractions

It was mentioned before, that an increase in the EMG amplitude may be caused by an increase in the force production of the muscle under test or by muscular fatigue. Therefore, an indication of fatigue based on the amplitude change is only possible, if situations with identical force production of the muscle are compared. This demand can be fulfilled using test contractions of predefined force levels. An appropriate method was developed and applied in field studies by Hägg et al. (1987). Thereby the occupational activity was interrupted at predefined moments of time and muscle contractions of known force were performed in a certain posture. EMGs were measured during these test contractions and the corresponding amplitudes were compared. Possible EMG changes over time observed under such conditions were attributed to the occurence of muscle fatigue. The use of this method needs an exact control of the body posture during the performance of the test contractions.

2.3. Selection of reference activities of comparable workload

In an analogous approach EMGs can be compared for selected periods of time which are characterized by a defined activity like holding or lifting a certain object, taking a defined body posture, or performing a special movement. The respective activities represent regular elements of the occupational work. They are taken as "reference activities" comprising a certain muscular force production. By contrast to the method described above, customary activities are used instead of test contractions and the interruption of work is avoided. When applying the reference task method activities should be preferred which are performed repeatedly during the execution of the work.

To meet this methodological approach a careful documentation of the sequence of operations is needed which enables, in the data evaluation process, a precise assignment of the subject's activity to the EMG. For this purpose a method for the production of an "electronic protocol" was developed and proven in practice (Ballé et al., 1982; Luttmann et al., 1988). A schematic representation of the concept and the equipment is shown in Fig. 1. The subject is accompanied during the total

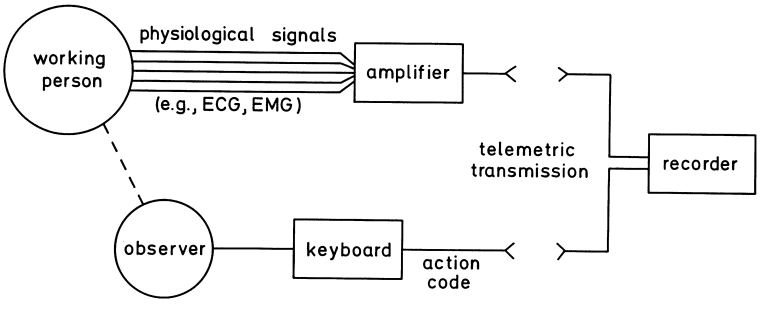


Fig. 1. Schematic description of the equipment used to record electrophysiological signals and an action code in ergonomic field studies.

working period by an observer who uses a numerical keyboard to encode the actual activity according to a predefined code list. The code list must be adopted appropriately for each field application. To this end actual characteristics of the work-load like body posture of the subject or mass and form of a manipulated object have to be considered. The electrical output signal of the keyboard represents an "action code" which is recorded together with the electromyograms and – if needed – other physiological signals like an electrocardiogram on the same recording device (computer memory media, magnetic tape). The coding system permits the recording of informations regarding the actual load at a given moment and to combine those load-related data with the current EMGs. In case of non-stationary workplaces it is favorable to use wireless transmitters to relay the action code as well as the physiological signals to the recording unit.

During off-line data evaluation, the electrical action code enables combining activity data with electromyographical findings and determining the myoelectrical activity during particular activities. The code signal can be applied in computer-assisted data evaluation in order to identify all periods during the shift connected with the same activity. With respect to the electromyographical fatigue analysis this method offers a valuable procedure to select sections from EMG time functions which are associated with similar workload and presumably equal muscular force production. For the fatigue indication, EMGs of all selected sections are compared, and a possible increase in the amplitude or a shift in the frequency spectrum towards lower values over time can be attributed to the occurance of muscular fatigue.

2.4. Joint analysis of EMG spectrum and amplitude (JASA)

In the aforementioned procedures for fatigue indication, EMG changes in the amplitude or spectral domain are considered separately and independently. Due to the twofold dependency of the EMG amplitude and spectrum on force and fatigue, the methods for fatigue indication are only applicable if EMG sections with identical force production are taken into account. This results in a considerable restriction for use in occupational field studies. In the newly developed method for "joint analysis of EMG spectrum and amplitude (JASA)" this restriction is reduced. For this purpose the methodology for fatigue indication was enhanced and both, the change in the amplitude and the spectrum were considered simultaneously (Luttmann et al., 1996a).

The JASA method is based on the principally known relationships between muscular force production and fatigue state, on the one hand, and the EMG amplitude and spectrum, on the other hand. These relationships were investigated in numerous studies; with regard to the EMG amplitude it is experimentally verified that the amplitude increases with increasing force as well as with the occurence of fatigue. A uniform functional relationship between the mechanical force production and the EMG amplitude is not entirely settled (Kumar 1996). The spectrum also depends on fatigue and force production: In case of fatigue a left shift in the spectral distribution was consistently found. For the force dependency of the spectral distribution, however, inconsistent findings were reported, depending on the muscle under test, the force level and the quantitative measure used for the description of the spectral distribution. The most often used measures to characterize EMG power spectral distribution are the Median Frequency (MF) according to Stulen and De Luca (1981) and the Mean Power Frequency (MPF) as defined by Kwatny et al. (1970). Measurements have been performed to study the MPF and MF as a function of the muscular force and torque level during both static and dynamic contractions (e.g. Komi and Viitasalo, 1976; Viitasalo and Komi, 1977,1978; Petrofsky, 1980; Petrofsky and Lind, 1980; Petrofsky et al., 1982; Hagberg and Ericson, 1982; Broman et al., 1985; Moritani and Muro, 1987; Nagata et al., 1990; Bilodeau et al., 1990,1991,1992a,b). A collation of literature data regarding the relationship between the force and both spectral measures is provided for studies on extremity muscles in Fig. 2. A relatively uniform result is achieved for MF. In most of the studies for various muscles and different types of contractions (step contraction, ramp contraction) an increase in

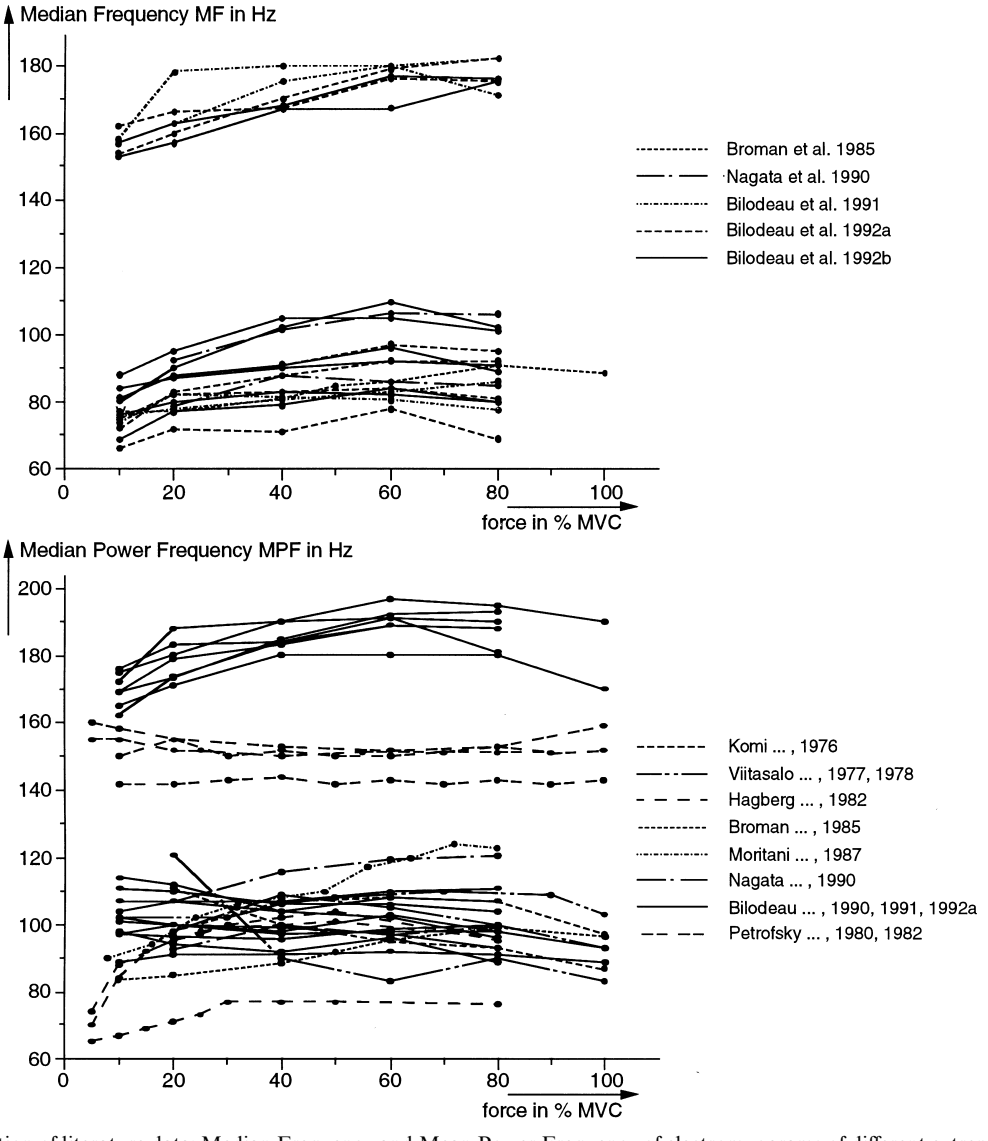


Fig. 2. Collation of literature data: Median Frequency and Mean Power Frequency of electromyograms of different extremity muscles as a function of the muscular force level expressed as the percentage of the force during maximum voluntary contraction (MVC).

MF is observed with increasing force up to forces of more than 60% of the maximum voluntary contraction force (MVC). For MPF also an increase is shown in the majority of the studies for force levels up to 30% and a levelling off at higher forces. In some cases even a decrease in MPF over the total force range was observed. In the following, due to its more uniform behavior the use of MF was preferred in the JASA method for the characterization of the dependency of the EMG spectrum on the force level.

The principle relationships between the EMG amplitude and spectrum, on the one hand, and the fatigue state and force production, on the other hand, are illustrated schematically in Fig. 3. The amplitude is assumed to be quantified by the

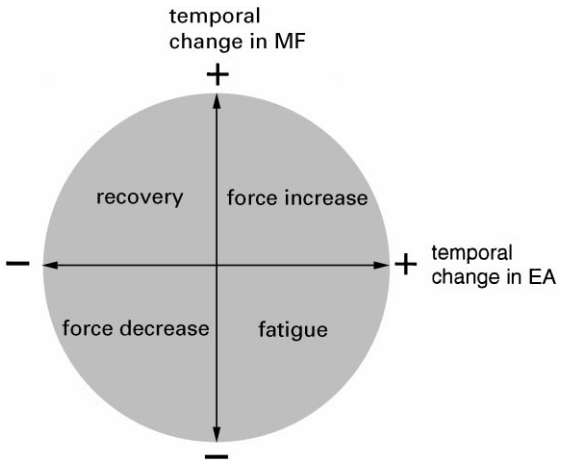


Fig. 3. Schematic representation of the method for the Joint Analysis of EMG Spectrum and Amplitude (JASA): Time-related changes in the Electrical Activity (EA) and the Median Frequency (MF) are considered jointly in order to differentiate between various EMG-change causations.

Electrical Activity (EA) which is derived from the raw EMG by rectification and low-pass filtering. Spectral distribution is characterized by MF. A temporal increase in EA – illustrated in the right-hand half-plane of the *xy* diagram in Fig. 3 – may result from an increase in the force production over time or from fatigue. A temporal decrease, shown in the left-hand half-plane, may be effected by force decrease or recovery from previous fatigue. If such a change in the amplitude is considered solely, the discrimination between a change in the fatigue state or in the force production is obviously not possible. A similar problem arises with respect to the spectrum: A left shift in the power spectral density indicated by a negative temporal change in MF – see the lower half-plane in Fig. 3 – may be received from fatigue or, according to Fig. 2, from decrease in force. Analogously, in the upper half plane an increase in MF may result from force increase or recovery. If both characteristics of the EMG are considered jointly, more reliable conclusions can be drawn than from the use of one of the characteristics only. Four cases can be distinguished: (i) For EMG recordings for which a simultaneous increase in EA and MF over time is observed, the concordant statement drawn from the change in amplitude and spectrum is an increase in muscle force (upper right-hand quadrant in the *xy* diagram in Fig. 3). (ii) An increase in EA which occurs together with a decrease in MF can be regarded as the result of muscle fatigue (lower right-hand quadrant). (iii) Similarly a decrease in EA and an increase in MF points to recovery from previous muscle fatigue (upper left-hand quadrant). (iv) A decrease in EA accompanied by a decrease in MF can be regarded as a sign of muscle force decrease (lower left-hand quadrant).

When applying the JASA procedure described above, knowledge about the temporal behaviour of the EMG amplitude and spectrum or measures thereof is needed. In previous studies (e.g. Luttmann and Jäger, 1992; Luttmann et al., 1989,1996a,b) such information were obtained from long-term EMG recordings by calculating mean values of EA and MF for succeeding short periods of time (e.g. 5 s or 10 s). This procedure results in time series for both characteristics with the respective sampling interval. EA and MF time series were summarized by regression analyses and the regression coefficients for EA and MF were regarded as quantitative indicators of the temporal change in the EMG amplitude and spectrum.

3. Examples for occupational fatigue analyses

3.1. EMG recording and evaluation

Electromyographical field studies using the aforementioned techniques for selecting reference activities as well as the JASA method were performed in various occupational settings during total shifts (e.g. manual materials handling in a weaving mill, price-recording at scanner checkouts, surgical work using endoscopic operation techniques). In all cases surface electromyograms from various muscles were recorded using a telemetric procedure. For this purpose electrodes were affixed to the skin by self-adhesive rings. After preamplification up to 5 EMG signals were transmitted via a telemetric system and recorded on an analog magnetic tape. In data evaluation from the raw EMG the Electrical Activity (EA) was formed by rectification and continuous averaging over

a time window of particular length (between 140 and 400 ms); the EMG spectra were calculated from the raw EMG by Fourier transformation, and the Median Frequency was determined in order to quantify the spectral distribution.

Parallel to the EMGs the action code describing the occupational activity of the person under test (cf. Section 2.3) was recorded on the same tape and used in data evaluation for the selection of reference activities representing EMG sections which are associated with similar muscular force production.

3.2. Fatigue indication using reference activities

The first example for an occupational fatigue analysis based on the evaluation of EMGs during reference activities is taken from a study on persons manipulating objects in a weaving mill. The task of the person under test was to produce warp beams. In the course of this activity bobbins with a mass of about 10 kg had to be transferred from transport containers into a special rack arrangement, called beamer. Grasping and lifting of the bobbins requires high muscular forces of the arm and finger flexors. Accordingly EMGs of the m. biceps brachii and the m. flexor digitorum superficialis of both arms were recorded and analysed. The activity of the person under test was documented during the work using the aforementioned coding system (see Fig. 1). During the study the following activities were encoded among others: lifting a bobbin, mounting a bobbin onto a peg in the beamer, and tying threads together.

For the purpose of fatigue analysis the lifting activity was chosen as a reference activity and the temporal behaviour of the EMG amplitude for such activity sections was followed up. The procedure will be explained using Fig. 4: Two short EMG sections, each lasting about 3 min, are shown. The time interval between the sections amounts to

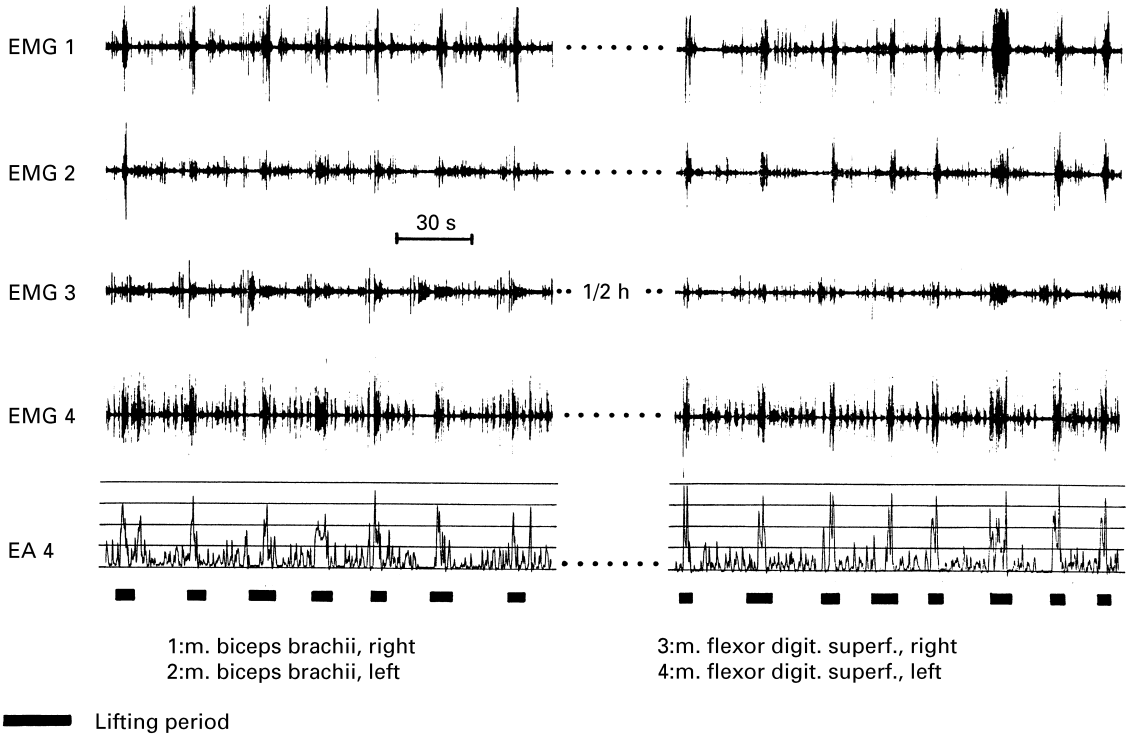


Fig. 4. Two sections of an original recording of the electromyograms (EMG) and Electrical Activity (EA) of the biceps and finger flexors during the handling of bobbins in a weaving mill, with a time lapse of about half an hour between the sections; black bars denote sections with lifting of bobbins.

about half an hour. Each lifting of a bobbin is indicated by a black bar underneath the EMG recordings. The point in time and the length of each lifting period were derived from the corresponding action code signal. Fig. 4 indicates that the highest EMG amplitudes occur during the lifting periods. Furthermore, it becomes evident from the EA recording (lowest trace of Fig. 4) that the peaks within the time curves are higher in the right section than in the left one. This points to an increase in the myoelectrical activity in the course of the work. The finding shown here for short sections only, is confirmed by a detailed analysis of all of the peak activities for total shifts (Luttmann and Jäger, 1992). It is concluded that the observed increase in the EMG peak amplitude during lifting indicates the occurence of muscular fatigue. The other possible explanation for the increase in the EMG amplitude, namely the increase in muscular force, is unlikely for this activity, since the mass and the form of the handled objects were unchanged and the lifting technique as well as the body posture did not vary systematically.

In Fig. 5 another example for the fatigue analysis based on the determination of the EMG amplitude for reference activities is shown. Data from a study of persons working on scanner check-outs in a supermarket (Luttmann et al., 1989) are provided. EMGs were derived from various shoulder muscles (m. trapezius, m. deltoideus) and arm muscles (m. biceps brachii, m. extensor carpi ulnaris) and recorded using the aforementioned telemetric system. Muscles from the left-hand side of the body were chosen, since, due to the construction of the cash-desk, almost all articles were moved using the left hand. The current activity of the subject was recorded by producing an action code according to Fig. 1. Additionally, an electrical scanner signal was recorded which was triggered by each movement of an article across the scanner in order to count the number of price recordings per unit of time. In contrast to the example shown in Fig. 4, where short sections of the EMG recordings are provided, here data for a whole working day are demonstrated.

In the upper traces of Fig. 5 each point refers to the average Electrical Activity in a time section relating to the action "scanning" (i.e. price

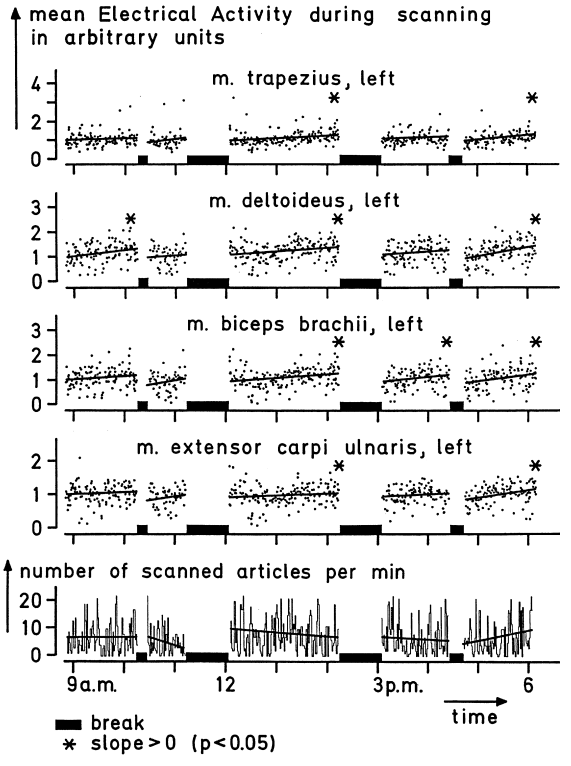


Fig. 5. Mean Electrical Activity of various shoulder and arm muscles during scanning (i.e. price recording by moving an article over an optical scanner) and scanning rate for a total working day at a check-out station in a supermarket. Time dependencies are represented by regression lines (straight lines) for 5 work sections.

recording by moving articles over the scanner). In the lowest trace, as a measure of the working speed, the number of scanned articles per minute is recorded. The time courses demonstrate that the work over the day is divided by breaks (see the black bars) into 5 activity sections. Within the sections the EA data and the number of articles are summarized by regression lines. A significant increase in the EA is found for at least one of the muscles for four of the five working sections. By contrast, the regression lines for the number of articles do not increase or decrease significantly over time; so the mean workload within the sections can be interpreted as not changing. It is therefore concluded that the measured increase in the mean EA is not the result of a change in load but an indication of muscle fatigue. The results shown in

Fig. 5 for one subject during a total work shift were confirmed by further measurements on 4 persons during total working days (Luttmann et al., 1989).

3.3. Fatigue indication using the JASA method

The method described in the previous section can only be applied, if an identification of reference activities comprising similar workload is possible. In a more general approach the newly developed JASA method for the joint analysis of EMG spectrum and amplitude is used in which, in addition to the change in the EMG amplitude, the change in the spectral distribution is considered.

An application of the method will be demonstrated referring to an example of an occupational EMG field study performed on surgeons in an operation theatre (Luttmann et al., 1996a). The EMGs from several muscles in the shoulder-arm region (right and left m. trapezius, right m. deltoideus) and the back (left m. erector spinae) were derived from surgeons during endoscopic operations in urology. In such operations a so-called resectoscope is introduced into the body through the urethra. The resectoscope mainly consists of a rod-shaped endoscope used for the visual inspection of the operation area and a wire loop supplied by a high frequency current for the dissection or coagulation of the tissue. The first part of the study was performed when the surgeons used the so-called "direct endoscopy". In this method, the surgeons looked "directly" into the urethra and the bladder through the endoscope. This method requires that both hands of the surgeons simultaneously grasp the instrument and that one eye is permanently in contact with the aperture of the endoscope. When positioning the instrument it is often necessary to steeply incline the trunk and to remain in this posture for a long time. A typical body posture during an operation using direct endoscopy is shown in the upper photograph in Fig. 6.

Data from the first part of the study were compared with the results of a second part which was performed after an ergonomic intervention. In this a video system was introduced enabling the surgeon to apply the "monitor endoscopy". For this purpose a camera was mounted on the top of the endoscope and the operation area could be visually inspected using a monitor. The monitor could be adjusted according to the anthropometry of the surgeon and the patient. Furthermore, a chair with arm and back rests was used which allows to support the back and the elbows of the surgeons from time to time in order to unburden the muscles. In the lower photograph in Fig. 6, the newly introduced equipment (Höcker et al., 1995) including the monitor hanging on the ceiling and being adjustable in the horizontal and vertical directions as well as the newly developed chair (Sökeland et al., 1995) is shown. Using this device the operation can be performed with the trunk held in an almost upright position.

In the first part of the investigation, the EMG studies were performed on four surgeons during 14 endoscopic operations lasting between 18 and 83 min. The second part was carried out more than one year after the ergonomic redesign of the operation equipment when the surgeons were accustomed to the new arrangement including the video technique. It was performed on 5 surgeons during 12 operations lasting between 18 and 80 min.

EMG amplitude was quantified by determining the Electrical Activity. Spectral distribution was analysed by performing Fourier transformation and calculating the Median Frequency. The temporal behaviour of both characteristics was evaluated as described before in Section 2.4 by computing the means for EA and MF for sections of 10 or 5 s length, respectively. For the resulting EA and MF time series a linear regression analysis was performed. The slopes of the regression lines for EA and MF are used in the further analysis as characteristics of the temporal behaviour of the EMG. According to the principle of the JASA approach (see Fig. 3) changes in the EMG were attached to one of the categories fatigue, recovery, force increase, or force decrease.

Results from the first part of the study during the application of direct endoscopy are provided in Table 1 for a total of 45 EMG recordings during 14 operations. (Data from 11 recordings could not be considered in data evaluation owing to disturbances induced by high frequency currents used for the dissection and coagulation of the tissue.) It reveals that fatigue occurs in the right trapezius in the majority of the operations (11 to 14). With

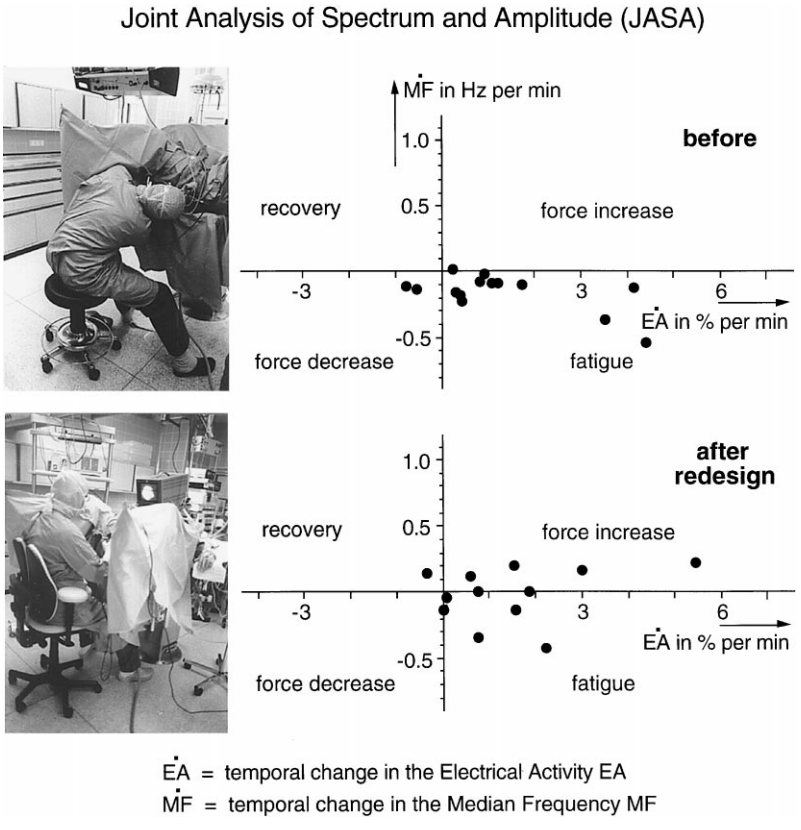


Fig. 6. Left-hand part: Photographs of typical work situations during an urological operation applying direct endoscopy before ergonomic redesign (above) and using monitor endoscopy after redesign (below). Right-hand part: Corresponding joint analysis of EMG spectrum and amplitude (JASA) for EMG recordings from the right m. trapezius of the surgeons.

Table 1
Time-related changes in the amplitude and spectrum of elctromyograms of various muscles derived from surgeons during urological operations applying "direct endoscopy"[a]

	Number of operations with simultaneous			
	EA increase and MF decrease	EA decrease and MF increase	EA increase and MF increase	EA decrease and MF decrease
Interpretation	Fatigue	Recovery	Force increase	Force decrease
M. trapezius, right	11	0	1	2
M. trapezius, left	4	1	3	1
M. deltoideus, right	4	2	3	0
M. Erector spinae, left	4	1	4	4

[a]EA: Electrical Activity; MF: Median Frequency.

respect to the other muscles fatigue was identified for 4 operations in all cases. It may concluded that the trapezius muscle forms a bottleneck for the performance of this activity. Signs of recovery were found altogether in 4 cases, only. The number of operations for which a change in the force

production was identified depends on the muscle, it varies between 4 (in case of the m. erector spinae) and 0 (m. deltoideus). It is concluded that during direct endoscopy fatigue occurred in the majority of the operations, the right trapezius muscle being particularly affected.

In Fig. 6 the influence of the ergonomic intervation on body posture and muscle fatigue is shown. In the left-hand part photographical represention of the typical pustures of the persons under test before and after redesign are shown. Before redesign, direct endoscopy is applied and a steep lateral bending of the surgeon can be seen; after redesign, when using the monitor method, the operation can be performed in an almost upright position. In the right-hand part of Fig. 6 the comparison of EMG findings from both parts of the study is provided. Data are focussed on the right trapezius muscle since this muscle was identified to form a bottleneck for the execution of the operation. In accordance with the schematical description of the JASA method presented in Fig. 3, the time-related change in EA is plotted on the x axis and the change in MF on the y axis. For each operation a pair of values was formed using the EA and MF regression coefficients and graphically presented in form of a dot in the xy diagram provided in the right-hand part of Fig. 6. Before redesign (upper diagram) about 80% of the EMGs are characterized by an increase in EA over time and a decrease in MF. The corresponding dots are located in the "fatigue quadrant" (lower right-hand quadrant in the diagram). After redesign (lower diagram) this amount was lowered to about 45%. It is concluded that for the muscle affected most, i.e. the right trapezius muscle, fatigue is reduced by the ergonomic intervention but not completely avoided.

4. Discussion

4.1. Methodological restrictions

EMG amplitude and spectrum depend not only on the fatigue state but also on the produced muscular force. Nevertheless, the separate use of the increase in the EMG amplitude or the left shift in the EMG spectrum for the indication of fatigue is possible, if the force production is similar for all EMG sections which are included in the analysis of the EMG time course. This has been proven by the use of test contractions (Hägg et al., 1987). The application of test contractions in occupational studies reveals some problems, however, since the body postures which are adopted during the performance of the test contraction have to be reproduced and controlled with high accuracy; otherwise fatigue-induced changes in the EMG might be covered by changes in the EMG caused by alterations in muscle geometry due to postural variations and changed muscular force production. Furthermore, in many real activities, such as the execution of the surgical work shown in the previous section, it is not possible to influence the content of the work by incorporating such test contractions.

Another possibility for selecting sections with similar force production is to use "reference contractions" which are a natural part of the normal work of the person under test. Here also, strictly speaking, only sections should be compared for which an identical force production is attained. Under real occupational conditions this demand can be fulfilled with some restrictions only. In field situations the same activity is normally performed somewhat diversely, e.g. with respect to the body posture, working velocity, and working technique. This results in a certain scattering of the EMG data for the same activity. Nevertheless, reliable informations regarding the EMG time course can be obtained and significant changes in EMG can be found, if the selected reference activities form an essential part of the work and are, in consequence, performed with a high degree of repetition. Then the number of EMG samples usable for the time course analysis is high and statistical methods such as regression analyses can be used in order to prove the significance of EMG changes as shown in the example provided in Fig. 5.

The indication of muscular fatigue is enhanced by applying the JASA method since changes in the EMG due to a change in the force production can be separated from fatigue-induced changes. In the JASA approach, muscular fatigue is assumed in cases, only, where an increase in EA as well as a decrease in MF over time was observed. In the

graphical representation of the JASA diagram (Fig. 6) the corresponding dots are positioned in the lower right-hand quadrant. It should be noted, however, that the locations of the dots within this quadrant scatter within a wide range. Some dots are located near the x or y axis, respectively, others are lying in an "intermediate field" between the axes. For dots near one of the axes a distinct change is observed in one of the characteristics EA or MF with only a slight change in the other parameter; for dots in the intermediate field both parameters are changing considerably over time. A systematic analysis of the causation for the different locations of the dots using the data shown in Fig. 6 is not advisable, since the respective EMGs were recorded under field conditions and some supposed influencing factors like body posture or body movements were not controlled with sufficient accuracy. Therefore, corresponding laboratory studies with fatiguing muscle contractions of constant force were performed. In such studies an influence of the size of the muscle and the individual force generating capacity of the muscle under the test on the individual reaction in the parameters EA and MF was observed (Luttmann et al., 1997,1998). Accordingly, the individual maximum force maybe partly responsible for the individual reaction. For further interpretations additional studies are necessary.

4.2. Physiological implications

The basis for the JASA method are experimentally proved relationships between EMG amplitude and spectrum and its measures EA and MF, on the one hand, and force and fatigue, on the other hand. Nevertheless, the appointment of a change in the phenomenological parameters EA and MF to the physiological causes is difficult, due to the complexity of the physiological processes.

An increase in muscular force is accompanied by an increase in EA. This mainly results from an increase in muscular activation, namely, of the number of action potentials per unit of time. It is performed by recruiting additional muscle fibres and/or by an increase in the firing rate ("rate coding"). The relative importance of the two neural mechanisms was described by Stein (1974). In muscle fatigue also an increase in EA is found. In this case two different factors contribute to the EA increase: (i) An increase in the activation of the muscle by recruitment and/or rate coding and (ii) a slowing in the propagation of the action potentials along the muscle fibres. Re (i) The fatigue-induced increase in the activation of the muscle occurs, since, as a result of fatigue, the force per action potential drops. (For review of the underlying physiological mechanisms see Luttmann (1996).) In order to maintain a constant force either additional motor units are recruited or the action potential firing rate is increased. Re (ii) Slowing of the action potential propagation velocity results from changes in the ional composition of the intra- and extra-cellular fluids in the muscle fibres and, in particular, in changes in the potassium distribution (Kössler et al., 1990; Sjøgaard, 1990). Such a decrease in the action potential propagation is reflected in a change in the EMG signal, in particular in a "broadening" of the bipolar action potentials measured on the skin surface. Both changes, the increase in the number of action potentials and their broadening, results in an increase in EA, since EA is determined by integrating the EMG and therefore, it depends upon the area below the EMG time functions. Another often applied integrative measure of the EMG amplitude is the Root Mean Square (RMS); its dependency on muscular force and fatigue is similar to EA.

In summary, integrative measures of the EMG amplitude such as EA and RMS are suitable to illustrate changes in the EMG signal caused by muscular fatigue; they are, however, not appropriate to solve the reverse problem, namely, to decide whether an increase in the EMG amplitude results from fatigue.

Another measure for the description of the EMG amplitude, the Muscular Activity (ACT), was defined some years ago by Spaepen et al. (1987). ACT is derived from the EMG not by integration, but by differentiation. It increases with increasing muscular activation and, in contrast to EA and RMS, it decreases with decreasing propagation velocity (Hermans, 1996). The simultaneous use of an integrative measure (EA or RMS) and a differentiating one (ACT) allows to distinguish between

force-related and fatigue-induced EMG changes. Based on this idea, a combination of both types of measures of the EMG amplitude, the ratio between RMS and ACT, was introduced recently as an indicator of muscular fatigue (Spaepen and Hermans, 1996).

An increase in muscular force is not only accompanied by an increase in the EMG amplitude but also by an increase in MF (see Fig. 2) which reflects a spectral shift to higher frequencies and is caused mainly by an increase in the number of action potentials per unit of time. Such behaviour of MF in relation to the firing rate is confirmed by model studies of Hermens et al. (1992). The change in the spectral distribution occurring during fatiguing contractions is influenced by counteracting effects: (i) A shift to higher frequencies due to the aforementioned increase in the number of action potentials and (ii) a shift towards lower values. In the literature this left shift in the spectrum has been discussed intensely for many years. (For review see Basmajian and De Luca (1985); De Luca (1985); Hägg (1992); Hägg and Kadefors (1996).) Two main interpretations have emerged: reduction in action potential conduction velocity and synchronization or grouping of motor unit firing. The reduction in conduction velocity is currently the most widely acknowledged interpretation of the fatigue-induced spectral change even if the decrement in spectral parameters is often greater than the decrement in conduction velocity (Broman et al., 1985; Merletti et al., 1992). However, there is some evidence that this explanation is only partially valid, in particular for low- and moderate-level endurance contractions (Krogh-Lund, 1993; Caffier et al., 1993). The net spectral change normally observed during fatiguing contractions is to the left; it is concluded that the spectral shift towards lower values exceeds the shift to higher ones.

One of the main interests in the fatigue analysis in occupational health and ergonomics is to define an indicator of the loss in the force generating capacity of the muscle under test. This information can then be applied for the justification of measures for work design like redesign of the workplace according to ergonomic principles or the introduction of a work-rest regimen which prevents immoderate fatigue. Another aim of fatigue analysis deals with musculoskeletal disorders and functional insufficiency. Static activity can be assumed to serve as a risk factor for the development of muscular pain symptoms (Westgaard et al., 1996), in particular for the trapezius muscle. In this muscle high static activity was observed, even if objects of low weight were handled (Strasser and Müller, 1999). The prevailing hypothesis for the occurence of shoulder myalgia is based on observations that low-threshold motor units are active throughout the contraction, until total relaxation (Kadefors et al., 1999). Therefore, the reduction of static activity by ergonomic work design and the introduction of breaks allowing muscular relaxation are assumed to be effective measures for the prevention of myalgia symptoms.

The aforementioned short compilation of the relationships between the EMG changes and the physiological processes indicates that all commonly used fatigue measures characterize changes in the myoelectrical activity, which coincide with fatigue in a complex way, but do not depend directly on mechanical characteristics. This elucidates the necessity of further development of an adequate measure of muscular fatigue which is preferably derived from surface EMGs and which depends directly on the interrelationship between the electrical EMG signal and the mechanical force development. A possible way for the derivation of such a measure is to determine an electrical equivalent of the "elementary force", i.e., the force which is produced by a single action potential, and its fatigue-induced reduction.

4.3. Conclusions regarding work design

An activity in which the working musculature is fatigued can only be performed for a limited period of time. After this time has elapsed, the musculature is exhausted and the needed force cannot be produced any longer. Another implication of fatigue is a loss in the required accuracy of movements. From a practical point of view, indication of muscular fatigue is needed in occupational studies in order to quantify the time for performing a certain activity without reaching the state of immoderate fatigue and disabling the person to continue the activity with the necessary precision. Such queries can be

studied using EMG findings, if the connection between fatigue-induced changes in the EMG and the endurance time of the muscle is taken into consideration. In the frequency domain, Hagberg (1981) has studied this connection by analysing the relationship between the temporal change in MPF and the endurance time. In the amplitude domain, the increase in EA caused by fatigue and the resulting endurance time was investigated by Laurig (1974,1975) using a fairly large number of experiments on different muscle groups. It reveals that a steep increase in the EA is connected with a small endurance time whereas a flat increase in EA is related to a large endurance time. The data were later extended by some further measurements (Laurig et al., 1987) and summarized using the regression formula $y = 10^{1.622 - 0.745 \log x}$ with $y =$ endurance time in min and $x =$ increase in Electrical Activity in percent per min (Luttmann, 1996).

For practical applications such relationships can be used to predict the endurance time for the measured EMG change. Investigations of this type were performed within the aforementioned studies on persons executing manual handling tasks in a weaving mill, during cash-desk activities and for surgeons performing endoscopic operations in urology. In all cases the periods of time for uninterrupted working sections were compared with the predicted endurance times, and if necessary, measures for the reduction in muscular strain and fatigue by introducing a new ergonomic work design or a physiologically justified work-rest regimen with shorter working periods could be derived and were recommended.

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